

# Varun Sharma

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## EDUCATION

### Bennett University

*Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence)*

Greater Noida, UP

*Expected May 2028*

- CGPA: 7.4/10.0
- Relevant Coursework: Data Structures & Algorithms, AI, Machine Learning, Linear Algebra, OOP

## TECHNICAL SKILLS

**Programming Languages:** Python (Proficient), C++ (DSA), TypeScript, JavaScript

**Web & Backend:** FastAPI, React, SQLAlchemy, Alembic, MongoDB, REST APIs, Google OAuth, Razorpay API

**Data Science & AI/ML:** NumPy, Pandas, Scikit-learn, Matplotlib, Google Gemini API

**Developer Tools:** Git, GitHub, GitHub Actions, Vercel, VS Code, Jupyter Notebooks

**Core Competencies:** Data Wrangling, Time-Series Analysis, Predictive Modeling, Backtesting, Risk Analytics

## TECHNICAL PROJECTS

### Algo Trade Simulator | *Python, FastAPI, React, TypeScript, MongoDB, Gemini API*

2026

[Live demo](#) · [Code](#) · Team project with Yashika Garg

- Built a full-stack backtesting platform that simulates trades on real market data with fees, slippage and no look-ahead bias, benchmarked against buy & hold and the S&P 500 on Sharpe, Sortino, drawdown and beta.
- Implemented walk-forward testing (tune on the past year, trade the next quarter on unseen data), which showed a tuned strategy's return falling from 24% to 11% a year out of sample.
- Developed a logistic-regression trading strategy from scratch with time-aware retraining, evaluated against a majority-class baseline and ROC-AUC.
- Integrated Gemini function calling so the AI copilot runs backtests and simulations; deployed on Vercel with Google OAuth and 107 automated tests in CI.

### RazorGrowth: Permissioned AI Revenue-Recovery Agent | *Python, FastAPI, SQLAlchemy, React, TypeScript*

2026

[Live demo](#) · [Code](#) · Razorpay AI Buildathon 2026 · Team project with Yashika Garg

- Built an AI agent that proposes Razorpay recovery campaigns for failed payments; LLM output is schema-validated and every proposal must pass a deterministic policy engine and explicit human approval.
- Engineered safe execution: atomic approvals, deterministic Razorpay reference IDs with retry, backoff and safe halt (no duplicate links), HMAC-verified webhooks and an audit trail.
- Designed a deterministic market simulator with a customer-behaviour model, an incremental-lift score and a fair weekly leaderboard.
- Shipped Google sign-in, Alembic migrations and 69 automated tests in CI; deployed on Vercel and FastAPI Cloud.

### AgriGuard: Satellite Imagery Crop Analysis | *Python, Scikit-learn, Sentinel-2 Data*

2026

- Developed a classification system to identify crop types and detect diseases using multispectral imagery.
- Implemented image preprocessing and feature extraction on large-scale Kaggle datasets.

### Sudoku Solver: Automated Logic Engine | *Python*

- Built an automated puzzle-solving application using a Recursive Backtracking algorithm.
- Optimized constraint logic for efficient solving of  $9 \times 9$  grids.

## LEADERSHIP & EXPERIENCE

### Management Lead

2025

*Sportikon 3.0*

### Peer Mentor

2025

*DSA Summit*

## HONORS & ACTIVITIES

- **Open Source:** Maintaining public repos for AI/ML and algorithmic research.
- **Hackathons:** Participant, Razorpay AI Buildathon 2026 (built RazorGrowth).